



Self-Organizing Jewelry Box Using Machine Vision

ME 3024 – Engineering Design & Economics

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CRN: 127559

Team Number: 17

DP3 – Design Phase 3 Technical Report

Members:

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Statement of Responsibility

Team 17 developed the concept of a self-organizing jewelry box that utilizes machine vision to identify and sort jewelry items such as rings, necklaces, and bracelets into compartments.

Elijah Freese –

Natalie McDonald –

Mariz –

Signatures:

_____EF_____ (33%) Elijah Freese

_____NM_____ (33%) Natalie McDonald

_____KM_____ (33%) Mariz Kedis

Table of Contents - -MARIZ

Table of Figures - -MARIZ

Introduction - -MARIZ

The Order of the Rings is a mechanized jewelry storage system that is the solution for the issue that students and young professionals face: the time spent managing unorganized and tangled jewelry in compartments. The present solutions for storage require total manual work, thus making the users annoyed by the mess and losing objects, and the chains getting tangled. In this report, we present every step of product creation made by Team 17 to come up with a small, mechanized gadget that mines images to identify and divide jewelry into separate boxes with 90% or more accuracy within 5 seconds or less per piece.

This document integrates the three phases of the design semester project. The report starts with project planning covering the opportunity identification process, customer profiling, and project management timeline. It continues with the findings of customer interviews and mapping customer needs to measurable engineering specifications by a Quality Function Deployment analysis. After the specifications, the report discusses intellectual property research, concept generation and selection, and the Failure Modes and Effects Analysis carried out to discover and control possible design risks. The last parts describe the 3D CAD model, physical prototype, Design for Assembly analysis, and Bill of Materials, ending with thoughts on the current prototype and suggestions for the next version of the product.

Project Planning - -MARIZ

Opportunity Exercise

At first, the team tried to identify the potential opportunities by coming up with various everyday problems that college students encounter, like untidy dorms, messy desks, and unorganized personal belongings. As a result of this activity, we identified organizing, especially jewelry, as the highest priority. Customers are hesitant mainly because they get tangled frequently. Finding the ring or earring is very time-consuming, and there is no smart or automatic solution available. Following the confirmation of several alternatives, the team finally decided on the automated jewelry organizer as the main topic for their semester project.

The project selected is for a new product platform. Although the market already has jewelry boxes, the existing products are entirely based on manual organization. Our system uses automation and machine vision features for differentiation, so we are not just upgrading an existing product, but creating a new product category. We, as a team, have decided on a tech leadership competitive strategy, as innovation and differentiation are far more important to us than cost competitiveness. By adding machine vision-based sorting, we are offering a feature that is not available in any other jewelry storage solution in the market.

The mission statement, shown in Figure 1, was developed by synthesizing the results of the opportunity to exercise customer frustrations and an assessment of technical feasibility. The team's goal was to create a product that is easy to use, reliable, compact, and affordable.

targeting college students and young adults as the primary market, with working professionals and collectors as secondary markets.

Mission statement	
Category	Details
Product Description	Automated jewelry storage system using machine vision to classify and sort items
Benefit Proposition	Reduces clutter, prevents tangling, saves time, improves accessibility
Key Business Goals	Prototype, ≥90% accuracy, affordable cost, scalable design
Primary Market	College students and young adults
Secondary Markets	Working professionals, older users, collectors
Assumptions	Users value convenience; vision works; users pay more
Stakeholders	Users, team, manufacturers, retailers

Figure 1, Mission Statement

Customer Profile and Use Case

During the planning phase, three potential customer groups were considered: The team chose these groups to cover a variety of organizational needs as well as different usage contexts. Firstly, college students were picked as the main target since their organizational problems are mainly due to small living areas and hectic schedules. Besides that, working women were brought in because they primarily look for efficiency and time savings. Most importantly, elderly women were considered in order to have a design that will be easy to use and accessible, especially for those users who have more delicate or sentimental items.

To identify the main target user, a customer profile was created. The face of our product in our minds is a fictional college student, Maxine Turner, 20 years old, studying biomedical engineering, and she is juggling studies, extracurricular activities, part-time work, and an active social life. Besides having a diverse jewelry collection that includes the heirloom necklace, Maxine's main considerations are convenience and self-expression. The profile is illustrated in Figure 2, and it served as the team's reference throughout the project and helped focus design on the real daily user.

CUSTOMER PROFILE

PROJECT: Automatic Jewelry Sorter

TARGET: College Students



Maxine Turner

Stats: 20-year-old woman, single, with significant other. Second year is Biomedical Engineering major that plays club soccer.

Attributes: She is a smart, studious person who balances school and extracurriculars on top of a part time job. She really enjoys her studies and wants to go pre-med.

Activities: Spending time with her friends, playing soccer, listening to music, and working out. Active member of her professional sorority is constantly out at different events.

Owns: Antique necklace passed down from her family, Ray Bans Aviator sunglasses, good clothing selection with a large variety, wide shoe collection to match different styles

Buying Habits: Keeps up to date with modern trends, looks for outfit ideas on social media and posts fit checks to her friends. Invests money in good quality items that can be reused for different looks.

Philosophy: Messy hair doesn't care. Socialite has good relationship with a lot of different organizations. Likes to be creative and express herself.

Figure 2, Customer Profile

Also, a use case was made to explain how a standard user would operate the system, as shown in Table 1. In the main scenario, a user places one or more jewelry pieces on the input tray. Then an image is taken, the kind of jewelry is determined by the machine vision algorithm, and the mechanical sorter is turned on to send the piece to the correctly labeled compartment. Another situation is when a single user wants to put it away quickly without getting mixed up with the other pieces. The use case proved that the design should not only reduce the physical user effort but also be capable of handling and recognizing various types of jewelry such as rings, necklaces, bracelets, and earrings, and have a neat storage method when not in use.

USE CASE

PROJECT: Automated Jewelry Sorter

TARGET: College Girl

USE CASE:

Putting away jewelry after a long day

Actors	Customer
Basic Flow	The customer has a lot of jewelry boxes sitting around with items that are all tangled together. The customer wants to organize all the items and consolidate it neatly so that they are easier to use. She places all of jewelry in sections on to the tray and waits for the machine to finish sorting before placing on the next load.
Alternative	The customer wants to quickly take care of jewelry without it getting lost in other items. She carefully takes of each piece and places it onto the tray to be added in with the rest of her collection.
Post Conditions	Sitting closed while charging on the vanity
Business Rules	The customer can use it for a wide assortment of jewelry as shown below: 

Table 1, Use Case

Project Management - -MARIZ

The team at Team 17 used a Gantt chart to plan and track the work of all project tasks from the start of the semester until the final presentation. As illustrated in Figure 3, the schedule lists 10 activities set from February to May 2026, including Needs Identification of Customers, setting Target Specifications, Reading of Intellectual Property, Concept

Generation, Concept Testing, Product Architecture, Design for Assembly, Economic Analysis, Prototyping, and Presentation of Final Results. The length of time for each task is shown by the bars on the right side, and the progress line, which is aligned with the deadline for DP3, May 6th, 2026, also gives an indication of the project status at the time of submission.

Essentially, the team stuck to the timeline laid down in DP1. The main reasons - identifying customer needs and establishing target specifications - were carried out punctually as the team gave highest priority to these two jobs at the start so that they could provide a basis for all other design decisions afterwards. The Gantt chart acted as a shared point of reference to make sure everyone in the team was aware of the time limits and the task dependencies. It really was a respite especially in the design and prototype phases when the load on everyone was maximum, it made them locate the places whereby they changed their efforts.

Reflecting on the project schedule, several tasks could have been executed in parallel to improve overall efficiency. Specifically, concept generation and the intellectual properties literature search overlap in their informational goals — insights from patent research directly inform concept ideas — and the team found that conducting these simultaneously rather than sequentially would have been beneficial. Similarly, early prototyping of subsystems such as the machine vision module and the mechanical sorting mechanism could have begun in parallel with the product architecture phase rather than waiting until both were fully defined. The team identified hardware integration, software debugging, and sensor calibration as the most likely sources of schedule risk, and time buffers were incorporated into the prototyping and testing phases to account for these possibilities.

The critical path for the project is identified to run through customer needs identification, target specification development, concept generation, and prototyping. Delay in any of these tasks would directly increase the overall duration of the project. Machine vision implementation and mechanical sorting design were on the list of major potential bottlenecks since working with reflective, variably shaped jewelry items is complex and achieving the 90% classification accuracy target requires several rounds of testing.

A Design Structure Matrix (DSM) was also developed to visualize the dependencies between project tasks. The DSM confirmed that specification development is a prerequisite for both concept generation and the QFD analysis, and that the economic analysis depends on outputs from the BOM and DFA tasks. Tasks with no upstream dependencies — such as the intellectual properties search — were identified as candidates for early parallel execution. The DSM analysis reinforced the team's decision to front-load customer-facing work early in the semester, ensuring that design decisions throughout the project remained grounded in verified customer needs.

Customer Needs - -MARIZ

In order to find out the needs of potential users, the team did six customer interviews with three main target groups, i.e. college students, middle-aged women, and elderly women.

Interviewees were picked from the team members' peers, family, and friends so as to be able to get a variety of different perspectives as well as to easily reach them. A brief summary of the interpreted customer needs is given in Table 2, and the full-length interview forms can be found in Appendix 1.

Customer Need	Importance	Priority
Needs to organize various types of jewelry	5	P
Doesn't damage any jewelry	5	P
Is small and doesn't take up a ton of room	3	S
Is aesthetic	3	S
Long cable / high battery life	4	P
Quiet when running	3	S
Is simple to use	4	P

Table 2, Customer Needs

Throughout the six interviews, some consistent needs kept coming up. The number one complaint that figured out by the majority of interviewees was necklaces and chains becoming tangled. They also said that the worst part is spending time untangling and even that might result in breaking of the delicate pieces. Therefore, first and foremost, we should ensure that tangle prevention becomes a key design feature. Another thing that came up several times was the wish for a quicker and more efficient way of organizing. For instance, college students pointed out that their tight schedules hardly allowed them time for manual jewelry management, in which case, automation would become a very highly appreciated feature.

Damage prevention was a strong concern across multiple interviewees, especially those with sentimental or high-value pieces, reinforcing the need for a gentle sorting mechanism. A unique and recurring suggestion was the ability to sort earring pairs together, raised by both Chloe and Lauren, which directly influenced the team's thinking on the granularity of the machine vision classification system. Cleaning capability was suggested by three interviewees but was designated as secondary given the added cost and complexity it would introduce to the prototype.

Notable variation was observed across customer types. College students prioritized speed, automation, and space efficiency. The middle-aged interviewee expressed skepticism about automation, suggesting the team focus marketing on users already frustrated with manual organization. The elderly interviewee emphasized simplicity and cost-sensitivity, indicating that ease of use and an accessible price point will be critical factors for that segment.

Specifications - -NATALIE

To effectively translate customer needs into measurable design targets, a Quality Function Deployment (QFD) matrix was constructed rating seven customer needs to nine technical specifications. These specifications were then validated by creating a comparison to competitor's products in a Competitive Benchmarking Analysis.

Customer needs were identified through interviewing a mix of audience members. Originally, we surveyed three types of audience members: college students, middle-aged working women, and elderly women to get a better understanding of who would be most likely to invest in our product. Ultimately, we found that college students are most likely to purchase and use our product, but surveying other audiences helped us expand our customer needs. The resulting highest priority needs are to organize various types of jewelry without creating any damage, reflecting our core objective and safety requirements of the product. Each specification in the QFD was assigned a direction of improvement: maximize, minimize, or target representing how we will strive to improve our specifications in the future. Specifications with strong relationships to multiple high-importance needs received the highest engineering priority in the matrix.

Correlation ratings between customer needs and specifications were assigned using values of 1, 3, and 5 representing weak, moderate, and strong relationships. Accuracy received a strong correlation with the need to organize various jewelry types, as the machine's vision classification system is the primary mechanism through which the system is sorting. Technical importance was calculated by multiplying each correlation rating by its corresponding customer need importance and summing across all needs per satisfaction. Presentable ranked highest at 15% of total technical importance, followed by accuracy at 14.2%. Technical difficulty was assessed on a scale of 1-5 reflecting the engineering challenging of achieving each specification, accuracy and damage tolerance were rated highest in difficulty as both require precise mechanical and software development.

	Importance (Customer Needs)	Accuracy	Time per item	Capacity	Size	Cost	Noise	Presentable	Lifespan / battery cycle	Damage Tolerance
		1	2	3	4	5	6	7	8	9
Direction of Improvement		↑	○	↓	↓	↑	○	○	↑	↓
CUSTOMER NEEDS:										
Needs to organize various types of jewelry	5	5	3							2
Doesn't damage any jewelry	5	3	2	1		2		1		5
Is small and doesn't take up a ton of room	3			2	5	2		3	2	
Is aesthetic	3	1		4	5	4	5	4	1	
Long cable / high battery life	4			3	3	4		2	5	
Quiet when running	3			2		2	5	4	4	2
Is simple to use	4	3	1					3		
TECHNICAL IMPORTANCE, DIFFICULTY, AND PRIORITY:										
Technical importance (calculated, absolute)		55	29	41	42	50	30	58	41	41
Technical importance (calculated, percentage)		14.2%	7.5%	10.6%	10.9%	12.9%	7.8%	15.0%	10.6%	10.6%
Technical importance (calculated, rank)		2	9	5	4	3	8	1	5	5
Units		dB	N	g	s	N/s/m	mm	mm	kN/m	kg
Marginal Target Value		> 10	480-800	< 3.5	< 13.0	0	33-50	37-45	> 65	< 1.4
Ideal Target Value		> 15	650-700	< 3.2	< 11.0	> 200	45	38	> 130	< 1.1
Technical Difficulty		5	4	3	3	2	1	4	2	5
Technical Priority (calculated)		275	116	123	126	100	30	232	82	205
Technical Priority (calculated, rank)		1	6	5	4	7	9	2	8	3

Table 1, DP1 QFD Matrix

Since there is no direct competing product currently in the market, four products with similar sorting functions were selected for benchmarking. The Stackers Classic Jewelry Box and Mele & Co. Empress Jewelry Box were chosen as baselines to represent the current market standard. Both devices use a manual sorting system, allowing use to establish a minimum standard for capacity and size. The Hero Pill Dispenser and Cassida C100 Coin Sorter were selected as baselines for an automatic system of similar functions. These two devices are small but have large capacities and can successfully sort varied physical items into categories, giving us a good gauge for specifications such as speed, accuracy, and noise.

Metric NO.	Needs Nos.	Metric	Imp.	Units	Dir.	Our Sorter (Target)	Stackers Classic Box (Manual)	Mele & Co Express Box (Manual)	Hero Pill Dispenser (Auto)	Cassida C100 Coin Sorter (Auto)
1	1,2	Accuracy	5	%	↑	>= 95	-----	-----	~90	~99
2	1,7	Time Per Item	4	sec/item	↓	<= 5	-----	-----	~3	0.24 coins/min
3	1,3	Capacity (# of categories sorted)	5	# of categories	↑	>= 10	-----	6	10	4 pennies/nickels
4	3,4	Size	3	in^2	↓	<= 100	69.9	183.6	~50	164.2
5	3,7	Cost	4	USD	↓	50	30 - 130	~80	\$30/month subscription	~180
6	6	Noise	3	dB	↓	<= 45	0	0	~60	-----
7	4	Aesthetics/ Quality	3	rating out of 5 stars	↑	>= 4	4.5	4.4	4.1	3.8
8	5	Lifespan / Battery Cycle	4	kN/m	↑	>= 130	-----	-----	Lifetime warranty	1
9	2	Damage Tolerance	5	N	○	<= 1.0	-----	-----	~0.5	~3.0

Table 2, Competitive Benchmarking Analysis

Marginal values were set just beyond the weakest relevant competitor for each metric, ensuring the product at minimum is more capable than the selected market solutions, and ideal values were anchored to the best-performing competitor in each category. For example, accuracy targets were set based on the Cassida C100, which has a near perfect accuracy sorting creating a string benchmark for our camera classification system. Noise targets were set to be lower than the Hero Pill Dispenser which many users describe as producing the same noise level as an espresso machine which would be unacceptable for a device that would most likely be kept in someone's bedroom. The damage tolerance metric was prioritized as a critical constraint rather than a performance goal as we want the device to be durable for moving, with the ideal value set at a gentle gravity-assisted handling force similar to the Hero' Dispensing system, since jewelry is much more delicate than coins or pills. Based on these findings, we reevaluated and finalized our product specifications as shown in Table 3 below.

Metric NO.	Need Nos.	Metric	Imp.	Units	Marginal Value	Ideal Value
1	1,2	Accuracy	5	%	> 88	> 99
2	1,7	Time per Item	4	sec/item	< 5	< 3
3	1,3	Capacity	5	categories	> 6	> 10
4	3,4	Size	3	in^2	< 165	< 70
5	3,7	Cost	4	USD	< 250	< 150
6	6	Noise	3	dB	< 60	< 40
7	4	Aesthetics	3	Star rating our of 5	> 3.8	> 4.5
8	5	Lifespan / Battery Cycle	4	kN/m	> 1	> 2
9	2	Damage Tolerance	5	N	<= 1	<= 05

Table 3, Finalized specifications after evaluating competing products

Economics – Revenue, Cost, Profit - -NATALIE

Our team decided to use Scenario 1 in selling our product, where we will sell the product at a fixed price per unit hoping that keeping it low and accessible will be more attractive to our target audience. Following this scenario we set these metrics:

Price: \$100/unit

Variable cost: \$50/unit

Fixed cost: \$10,000/year

Estimated sales: 300 units/year

Initially during design phase 1 we priced the units at \$60 per unit, but after doing the competitive benchmarking analysis the selling price was increased to better align with similar automated consumer devices available on the market while remaining accessible to our target market. Resultantly the product was analyzed under Scenario 1, in which the device is sold at a fixed price of \$100 per unit, which an assumed \$10,000 per year for fixed costs and a variable cost of \$50 per unit estimatred on projected materials and assembly costs. With these parameters the breakeven point was calculated at about 200 units and at our estimated sales volume of 300 units, total revenue is projected at \$30,000, total cost \$25,000, and profit at \$5,000. Selling 300 units places the product well past breakeven and demonstrates the

financial viability of the product at this price point. We would ideally like to make the cost lower so that it is more approachable for our target audience, but to sustain profit \$100 per unit puts us on par with other automatic products in the market while staying within reach of college students.

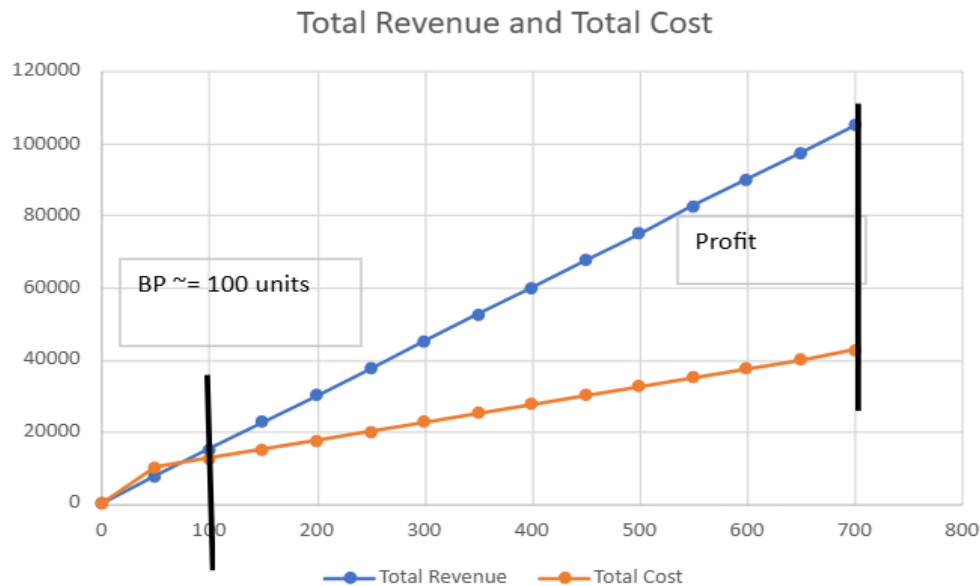


Figure 1, Total Cost vs Total Revenue

Total revenue and revenue were calculated by multiplying the selling price by the number of units sold, and total cost was calculated by adding the fixed costs to the product of variable costs and units sold. Furthermore, profit was then calculated by subtracting total cost from total revenue and breakeven points were identified by dividing the fixed costs by the difference between selling price and variable cost. These values were plotted across a range of 0 to 700 units to visualize where the product begins to produce profit as shown in Figure 1 above.

Intellectual Property - -NATALIE

In order to gain ideas for our system and understand any design constraints, a patent analysis was conducted on five different inventions. Because no patents exist for an automated jewelry sorting device specifically, patents were selected based on functional similarity to our design. US-12337353 and US-10766711 were selected for their relevance to automated sorting systems and control logic, US-11033936 was selected as the closest resemblance to our product concept, US-9035210 was selected for its optical sorting pipeline, and US-20220355980 was included for its small item organization and compartment design for small objects. Combined, these patents span the mechanical and software dimensions of our product and provided inspiration towards our working design.

Patent Number, Title, and Issue Date	Key Features of this Invention	Impact of this Patent on Current Design
US-12337353 Systems and Methods for Dynamic Sortation 2025-06-24	This patent describes an automated shuttle sorter where a robotic arm picks objects from an input area places them into a moveable carriage that travels along a track A scanner with several camera then identifies objects as they fall through the bin Bins are not preassigned but are assigned in real time based on object characteristics	Limitations on the team's design: • None Ideas gained from this design: • Bins do not need to be pre-assigned; the system can allocate sorting compartments in real time • Placing frequently sorted items closer to the home position could minimize travel time • Using multiple camera angles simultaneously to identify an object would improve classification accuracy.
US-20220355980 Bead Workstation and Magnetic Devices for Sorting Beads 2022-10-10	A magnetic bead sorter and workstation that uses magnets to secure small sorting cups to a magnetic board The product allows users to add as many cups as needed for projects requiring a lot of different bead colors, cups come in different colors and sizes to distinguish bead types	Limitations on team's design: • None Ideas gained from this design: • Soft foam interior surface would cushion jewelry upon placement • Lip design around compartment edges to prevent items from flipping or rolling out • Modular moveable compartments rather than fixed cavities

US-11033936-B2 Method and apparatus for sorting pieces 2021-06-15	This patent describes a consumer grade automatic sorter designed for LEGO bricks that uses a rotating helical screw tube to separate and propel small pieces one at a time System identifies pieces by color, size, and transparency and the rotating spigot system diverts pieces into the correct bin	Limitations: - Patent has been granted so the helical screw propulsion mechanism is protected and cannot be used directly without licensing Ideas for our design: - Using a neural network embedded on a low-cost microcontroller for real-time classification - Implementing a small vibration element to our input tray could help untangle jewelry
US-9035210-B1 Optical robotic sorting method and apparatus 2015-05-19	An optical robotic sorting system designed for sorting ears of corn on a conveyor belt. It uses LED illumination, a camera imaging device, and vision software to identify defective products then activates a robotic arm with a silicone vacuum cup to place items into sorting areas	Limitations: - Patent is granted so our gantry system needs to be carefully distinguished from this idea Ideas gained from this design: - Illuminate -> image -> analyze -> sort workflow - Vacuum cup design to move pieces instead of a gripping system - Building in a fault detection mode to stop the gantry is an item if it gets jammed or misidentified
US-10766711-B2 Modular package sorting system 2020-09-08	Modular package sorting system for commercial distribution centers where packages are scanned, weighed, and dimensioned, then assigned a sort of designation and passed with a data packet traveling alongside each package. Each module decides whether to pass the package forward or divert it to the next bin	Limitations - Patent is granted and active, and therefore cannot be directly replicated in design Ideas gained from this design: - Storing classification data alongside each jewelry piece as it moves through the gantry - Adaptive speed control based on size - Bin fill tracking and user notification system

		Flagging system for pieces that cannot be directly identified
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Table 4, Patent analysis table

Concept Generation and Selection - -NATALIE

After doing research and compiling all our specifications, we began to develop potential design solutions for our automatic jewelry sorter. Each team member independently generated two concept sketches before sharing and discussing as a group, resulting in 6 total design ideas. These concepts were then organized into a Concept Combination Table, which broke the design into six key subfunctions: power source, actuator, motor, number of axes, delivery method, capacity, and sensor type. Concept lines were then drawn through the chart connecting one option from each subfunction to define complete concepts, resulting in six unique design combinations represented by colored lines in the Figure below.

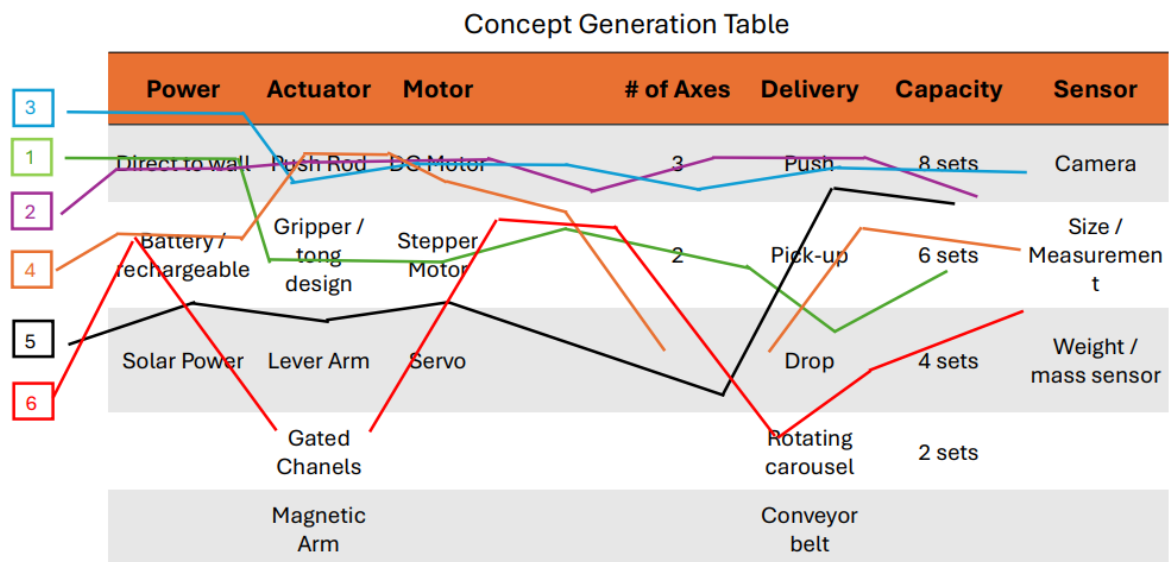


Figure 2, Completed Concept Generation Table

Following the gallery method presentation, the team received several written feedback from our peers. Some reviewers commented how they liked the inclusion of the number of axes as a subfunction category, noting it as thoughtful and a unique consideration. The solar powered option was also noted as a creative choice, though reviews acknowledged its less than ideal practical limitations considering it is an indoor device. However, reviewers commented that adding labels to the subfunction images and concept sketches would help improve clarity. We were also able to gain valuable ideas that could contribute to our design as one reviewer mentioned that combining all three sensor types would allow the device to better adapt to the variety of jewelry types and shapes. We were also recommended to further review the trade-offs between battery power and wall power with respect to profitability and desk positioning.

Using the feedback from the gallery review and team discussion, we evaluated the six concepts against established customer needs. The original concept sketches are shown below:

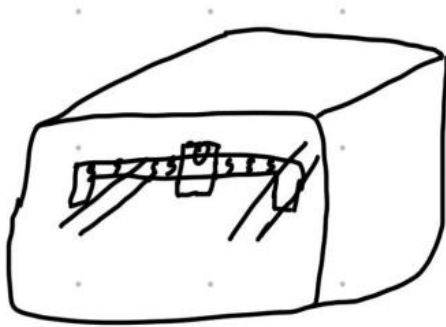


Figure 3, Concept 1

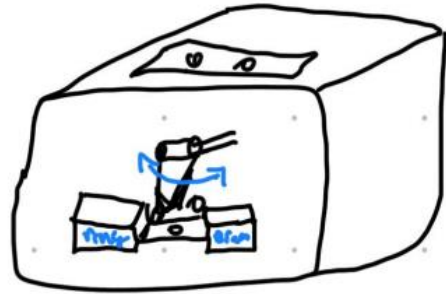


Figure 4, Concept 2



Figure 5, Concept 3

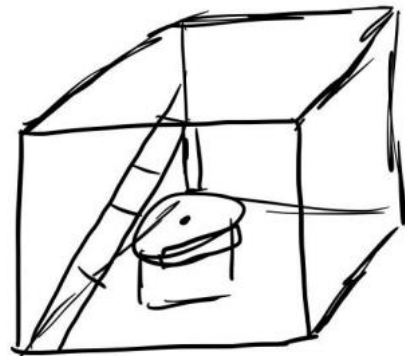


Figure 6, Concept 4

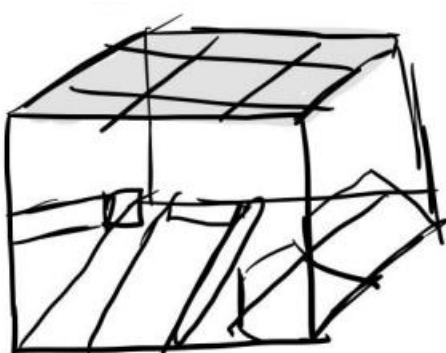


Figure 7, Concept 5

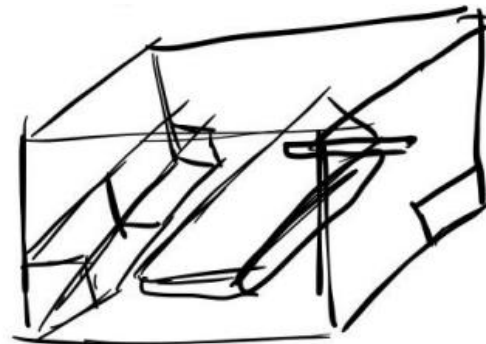


Figure 8, Concept 6

Concept #	1	2	3	4	5	6
Sub systems	Direct to wall Gripper Stepper motor 2 Axes Drop 6 sets Size/measurement	Direct to wall Push rod DC Motor 3 axes Push rod 8 sets camera	Direct to wall Push rod DC Motor 3 axes Push rod 8 sets size/measurement	Battery Push rod DC motor 2 axes Drop 6 sets size	Solar power Lever arm Servo 2 axes Push 8 sets weight	Battery Gated channels Servo 2 axes Rotating Carousel 4 sets weight

Table 5, Concepts list

The two axes gantry system emerged as the preferred architecture as it provides the most precise positional control, critical for gently picking up and placing delicate jewelry without causing any damage. A camera-based vision system that will sort based on size was selected as the primary sensor for classification, maintaining the level of accuracy established in our QFD analysis. In consideration of our user feedback, we also hope to include weight sensing as a secondary input to be used for classification in future improvement. Wall power was selected as the primary power source given that the device is intended for desktop home use where portability is less critical than a stable power supply. The gripper actuator design was selected over push rod due to its ability to handle irregular shapes gently, and we wanted to use a stepper motor so that we have precise control over the movement of the gantry system. Therefore, our final concept selection combines two-axis gantry, a stepper motor, camera and size sensors, a gripping mechanism, and direct wall power.

Failure Modes and Effects Analysis (FMEA) - -NATALIE

FMEA analysis identified ten potential points of failure across the major subsystems of the automated jewelry sorter. The highest RPN scores were assigned to the input tray feeding multiple items simultaneously (RPN = 210) and camera misclassification (RPN = 168), indicating these are the two highest priorities. To mitigate multiple item feeding, the item input tray will incorporate a vibration mechanism and geometric constraints to encourage single item presentation to the camera, consistent with the approach seen throughout our research. Camera misclassification will be addressed through controlled dual-background LED lighting and a minimum confidence threshold. Gripper damage to jewelry (RPN) = 108) and output compartment overfilling (RPN = 144) were also flagged as significant risks, to be addressed through adjustable grip force limits with soft-tipped contacts and a fill-level sensor with a user notification system. The lower RPN items such as power loss and motor noise are considered manageable through standard controls.

Component	Potential Failure Mode	Potential Effect	Severity (1-10)	Potential Cause	Occurrence (1-10)	Potential Controls	Detection (1-10)	RPN
Camera Vision System	Misclassifies Jewelry type	Item sorted into wrong compartment	7	Poor lighting, reflective surfaces,	6	Background lighting, confidence threshold	4	168

				tangled items				
Gantry (X/Y) axes	Axes fail to move to correct position	Item dropped in wrong location or not picked up	8	Motor failure, belt slip, software error	3	Limit switches, encoder feedback	3	72
Gripper end effector	Fails to grip item	Item not picked up, system stalls	7	Item too heavy, irregular shape, gripper wear	5	Grip sensor feedback	4	140
	Grips too rigid	Jewelry damaged	9	Incorrect grip force	4	Adjustable grip force limit, soft material	3	108
Input tray	Multiple items fed at once	Two items classified as one	6	User dumps too many items, tangled jewelry	7	Vibration mechanism, geometry constraints	5	210
Power supply	Power loss during operation	Gantry stops mid-cycle, item dropped	8	Wall outlet issue, cable failure	2	Status indicator light	3	48
Classification Software	System cannot identify item	Item not sorted, sent to reject tray	5	Unusual jewelry type not programmed with	5	Reject/unknown bin, user notification	2	50
Output compartments	Compartment overfills	Items pile up, fall out, get damaged	6	User doesn't empty trays	6	Fill level sensor, app notification	2	144
Stepper Motor	Excessive noise during operation	Violated noise specification	4	Motor vibration, loose components	4	Dampening mounts, careful stepper motor selection	5	80
	Overheats during extended use	System shuts down, potential damage	7	Continuous operation without reset	3	Thermal cutoff, duty cycle limits	4	84

Table 6, FMEA

3D CAD Model and Prototype -ELIJAH

Design for Assembly -ELIJAH

Bill of Materials / Design for Manufacturing -ELIJAH

Reflections on Physical Prototype and Product Improvements -ELIJAH

Conclusion - -ELIJAH